M2 SETI
Quiz B4 Embedded Linux Embarqué

First Name:
Family Name:

Instructions
- Duration: 2 hours
- Any communication during the quiz is strictly forbidden
- A sheet of A4 paper with what you want on it is allowed
- Any other document, course material, TP material... is forbidden
- Calculating devices (calculator, computer, cell phone, etc.) are forbidden
- Please write as legibly as possible, any illegible answer will be considered as wrong
- Do not forget to write your name and surname on your copies and to return the subject at the end of the the exam

Scale
1 point per question, no penalty for multiple choice questions. Q21 holds 5 marks.

Questions
Question 1 : Name and detail three advantages of using Linux on an embedded system.
Question 2: Name three actions that the bootloader must perform in order to prepare the execution of the Linux kernel.
Question 3: Explain why general purpose Linux distributions use an initial memory image (initramfs) at Linux kernel boot time.
Question 4: Indicate what the init process corresponds to. Name two of its roles.
Question 5: Explain what a cross-compiler is and why it is often used for embedded systems development.
Question 6: Name four directories that are typically found at the root of the root file system of a Linux system. Briefly indicate, for each of the directories mentioned, what it is used for.
Question 7: Explain what is the purpose of the buildroot tool used in the lab.
Question 8: Name four major differences between the development of an application in classical C and the development of code in C for the Linux kernel.
Question 9: Explain what a kernel module is.
Question 10: When a module (named foo) is started, the kernel will (only one correct answer, no justification required):
1. Call the main function of the module
2. Call the function foo_init of the module
3. Call the function that has been declared by the module_init macro
4. None of the previous answers is correct

Question 11: Indicate why, from the point of view of the discovery and identification of the peripherals connected by the kernel, the USB and I2C buses are different.

Question 12: Explain how the discovery and matching between a device and the driver is done in the case of the I2C bus.
Question 13: What happens when a device, which the foo driver knows how to manage, is detected by the kernel (only one correct answer, no justification required):
1. The kernel calls the main function of the foo driver
2. The kernel calls the initialization function of the foo driver
3. The kernel calls the foo_probe function of the foo driver
4. The kernel calls the callback function stored in the probe field of the structure describing the driver
5. None of the above answers is correct
Question 14: Explain how a user-space application interacts with a driver of a character mode device.
Question 15: A device driver can transfer data to/from an application running in user space (one or more correct answers, no justification required):
1. Using a simple dereference of a pointer pointing to user space
2. Using a classic memory copy function (memcpy)
3. Using the copy_to_user and copy_from_user functions
4. None of the above answers is correct
Question 16: What happens when a driver calls the wait_event function in response to a request made by an application (a read for example). Indicate when this function is terminated (i.e., when the driver terminate (in other words, when the instruction following the call to this function begins to execute).
Question 17: Which of the following are grouped into directories and subdirectories?
a) register
b) memory
c) files
d) routines
Question 18: Which character is known as a root directory?
a) ^
b) &
c) &&
d) /
Question 19: How an embedded system communicate with the outside world?
a) Memory
b) Output
c) Peripherals
d) Input
Question 20: Which of the following unit protects the memory?
a) memory management unit
b) peripheral unit
c) execution unit
d) bus interface unit
Question 21: Code a set of 10 instructions with the following scheduling set (called by the kernel):
a) running
b) sleep

c) interrupt and uninterrupt
d) kill
e) daemon